
Deepfake Detection: Analysis of Deepfake Detection Using Multiple Models

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Abstract

The proliferation of deepfake technology poses serious threats to privacy, security, and information authenticity. This study examines the performance of several deepfake detection models, including ResNet, EfficientNet, Vision Transformers, and Deepfake Defender, using the DeepFake Face dataset for fine-tuning. Generated with advanced diffusion models, the dataset focuses on face images to enable precise model evaluation. We further assess the generalizability of these models on the DiffusionDB-Face dataset, which also comprises diffusion-based deepfake face images. The results highlight the strengths and limitations of each model, providing benchmarks and insights for advancing deepfake detection technologies.

1 Introduction

The rapid advancement of deep learning techniques has revolutionized the generation of hyper-realistic synthetic media, commonly known as deepfakes. These AI-generated facial images and videos pose significant threats to privacy, security, and information integrity. The malicious use of deepfakes in misinformation campaigns, identity theft, and defamation has eroded public trust in digital content, making deepfake detection a critical research area.

Traditional detection methods face increasing challenges due to the continuous improvement of deepfake generation technologies. Diffusion models, one of the most advanced techniques for generating synthetic images, have proven particularly difficult for existing detectors to identify accurately [1]. To address these challenges, this study investigates the effectiveness of various state-of-the-art detection models, including ResNet, EfficientNet, Vision Transformers (ViT), and the composite model DeepFake Defenders, in identifying deepfakes generated by diffusion models.

Our approach leverages two recently developed datasets: the *DeepFakeFace (DFF)* dataset and the *DiffusionDB-Face (DFDB)* dataset. Both datasets focus on face images generated using advanced diffusion models, providing a comprehensive benchmark for evaluating model performance. By fine-tuning the detection models on the DFF dataset and testing their generalizability on the DFDB dataset, we aim to enhance the robustness of these models in detecting high-quality synthetic images.

This work provides a comparative analysis of different detection architectures, focusing on their ability to handle the challenges posed by diffusion-based deepfake images. The findings contribute to the development of more effective deepfake detection systems, enhancing digital security and mitigating the risks associated with synthetic media.

1.1 Datasets

The *DeepFakeFace (DFF)* dataset [1] is utilized for model training and evaluation. This dataset comprises 120,000 facial images, including both authentic and AI-generated images produced by diffusion models. It maintains the directory structure of the IMDB-WIKI dataset from which real images are sourced, making it a robust benchmark for fine-tuning deepfake detection models.

To assess model generalizability, we use the *DiffusionDB-Face (DFDB)* dataset [2]. This dataset includes 24,794 high-quality fake images generated by advanced diffusion models, alongside diverse text descriptions and metadata. DFDB is specifically designed to align with the latest advancements in generative AI, providing a diverse and realistic set of face images for testing the robustness of deepfake detection methods.

2 Related Work

The rapid evolution of generative AI models, such as Stable Diffusion and MidJourney, has heightened concerns about the misuse of deepfake technology. Detecting fake images and videos is increasingly challenging due to the growing accessibility and sophistication of these tools. As generative models advance, addressing their security implications remains a critical area of research.

Deep-learning-based detection methods have emerged as the most effective countermeasure against sophisticated generative models. We categorize classic deepfake detection approaches into three main structures:

CNN-based Models Convolutional Neural Networks (CNNs) have shown exceptional performance in image classification, including deepfake detection. ResNet, a widely used CNN, has been effective in detecting synthetic images when trained on specialized datasets [3; 6]. For instance, Wang et al. [4] demonstrated ResNet50’s ability to detect ProGAN-generated images by identifying systematic flaws in synthetic data. EfficientNet, another CNN-based model, has also achieved high accuracy, with Tan et al. [5] reporting an AUC-ROC score of 0.9974 in the DeepFake Detection Challenge (DFDC).

Transformer Models Vision Transformers (ViTs) have become increasingly popular for image analysis, often combined with CNNs to utilize their complementary strengths. Coccomini et al. [7] achieved near state-of-the-art performance on the DFDC dataset by integrating EfficientNetB7 with ViT. Similarly, Heo et al. [8] demonstrated that combining local and global features using vector-concatenated CNN outputs with ViT outperformed previous models on the DFDC dataset.

Diffusion Models Diffusion models have emerged as a leading method for high-quality image generation, raising new challenges for deepfake detection. Many existing detectors struggle with diffusion-generated images. Wang et al. [9] proposed Diffusion Reconstruction Error (DIRE), leveraging the distinct reconstruction behaviors of real and synthetic images in diffusion models to achieve reliable detection.

Despite advancements in detection techniques, the rapid development of generative models continues to challenge the generalizability of detectors. This study aims to address these challenges by analyzing datasets generated by the latest diffusion models and evaluating the performance and robustness of various detection architectures.

3 Fine-tuning Models

We fine-tuned multiple state-of-the-art models for image classification using the DeepFakeFace (DFF) dataset, which includes both real and diffusion-generated deepfake images. The evaluation metrics includes: **Accuracy (ACC)**, **Area Under the Curve (AUC)**, and the **Equal Error Rate (EER)**.

3.1 ResNet

ResNet, a classic CNN architecture, is known for its efficient parameter utilization and strong performance on image classification tasks. We fine-tuned ResNet50, ResNet101, and ResNet152

models to detect deepfakes in the DFF dataset. Overfitting was a key challenge, requiring careful tuning of the learning rate. A learning rate of 1×10^{-5} was selected to balance convergence and generalization.

Table 1 summarizes the results, showing high accuracy for images generated by Stable Diffusion v1.5, which achieved the best classification performance. Overall accuracy across the dataset reached 90.22%.

Table 1: Fine-tuning ResNet Models. SD:Stable Diffusion. Loss Function: Cross Entropy; Optimizer: Adam.

Model	Method	ACC	AUC	EER
ResNet50	SD v1.5	0.9821	0.9767	0.0126
	SD Inpainting	0.9397	0.9277	0.0481
	InsightFace	0.9101	0.8757	0.0555
	Real	0.9014	0.8637	0.0616
ResNet101	SD v1.5	0.9868	0.9842	0.0105
	SD Inpainting	0.9566	0.9438	0.0304
	InsightFace	0.9133	0.8854	0.0588
	Real	0.9104	0.8756	0.0555
ResNet152	SD v1.5	0.9883	0.9874	0.0108
	SD Inpainting	0.9653	0.9504	0.0193
	InsightFace	0.9195	0.8947	0.0558
	Real	0.9159	0.8859	0.0547

3.2 EfficientNet

EfficientNet [5] is a convolutional neural network optimized for high performance with fewer parameters and lower computational costs. Its core innovation, Compound Scaling, proportionally scales depth, width, and resolution to achieve an effective trade-off between accuracy and resource efficiency.

EfficientNet models (B0–B7) were fine-tuned for deepfake detection tasks. Initial attempts using TensorFlow focused on freezing most convolutional layers while fine-tuning only the fully connected layers. The training pipeline included data augmentation, the Adam optimizer, mixed precision training, class weights to address imbalance, and callbacks like EarlyStopping and ReduceLROnPlateau. However, this approach faced significant challenges, such as overfitting to the training data and poor generalization to the validation set. Freezing convolutional layers also limited the model’s ability to learn deepfake-specific features, and the low initial learning rate of 1×10^{-5} further slowed convergence.

To overcome these limitations, the fine-tuning approach was revised in PyTorch. A higher learning rate of 1×10^{-4} was used with dynamic adjustment to stabilize training. Selective unfreezing of higher convolutional layers allowed the model to capture more task-specific patterns. Gradient accumulation and mixed precision training ensured computational efficiency, while Dropout and L2 regularization were introduced to improve generalization.

These modifications significantly improved performance, addressing overfitting and convergence issues. The fine-tuned model’s results, including accuracy (ACC), AUC, and EER metrics, are summarized in Table 2.

3.3 Vision Transformer

Vision Transformers (ViTs) [10] adapt Transformer architecture to vision tasks by processing images as sequences of patches, effectively capturing global and local features. We fine-tuned a pre-trained vit-base-patch16-224-in21k model on the DFF dataset.

Key training configurations included a learning rate of 5×10^{-5} , data augmentation, and early stopping. The ViT model achieved an overall accuracy of 91.56%, with high performance across most image classes. Table 3 provides detailed of the finetuning results.

Table 2: Training results for EfficientNet models on deepfake detection. (Four classes: SD v1.5, SD inpainting, InsightFace, Real.)

EfficientNet-B0 to B3					EfficientNet-B4 to B7				
Model	Method	ACC	AUC	EER	Model	Method	ACC	AUC	EER
B0	SD v1.5	0.9899	0.9855	0.0058	B4	SD v1.5	0.9921	0.9917	0.0076
	SD Inpainting	0.9746	0.9630	0.0135		SD Inpainting	0.9898	0.9846	0.0048
	InsightFace	0.9444	0.9176	0.0289		InsightFace	0.9907	0.9881	0.0067
	Real	0.9392	0.9313	0.0530		Real	0.9914	0.9877	0.0048
B1	SD v1.5	0.9888	0.9894	0.0118	B5	SD v1.5	0.9950	0.9940	0.0040
	SD Inpainting	0.9785	0.9671	0.0098		SD Inpainting	0.9941	0.9913	0.0030
	InsightFace	0.9537	0.9435	0.0361		InsightFace	0.9949	0.9937	0.0038
	Real	0.9497	0.9274	0.0284		Real	0.9952	0.9933	0.0030
B2	SD v1.5	0.9940	0.9944	0.0064	B6	SD v1.5	0.9944	0.9923	0.0035
	SD Inpainting	0.9935	0.9892	0.0021		SD Inpainting	0.9938	0.9922	0.0045
	InsightFace	0.9992	0.9990	0.0007		InsightFace	0.9992	0.9988	0.0004
	Real	0.9994	0.9990	0.0001		Real	0.9997	0.9996	0.0001
B3	SD v1.5	0.9918	0.9904	0.0069	B7	SD v1.5	0.9954	0.9946	0.0037
	SD Inpainting	0.9891	0.9867	0.0086		SD Inpainting	0.9949	0.9924	0.0025
	InsightFace	0.9749	0.9684	0.0185		InsightFace	0.9952	0.9957	0.0053
	Real	0.9741	0.9609	0.0128		Real	0.9953	0.9918	0.0013

Table 3: Performance of Vision Transformer on DeepFakeFace Test Set

Overall Performance		Class-wise Performance			
Metric	Value	Class	Accuracy	AUC	EER
Accuracy	91.558%	Inpainting	0.9789	0.9928	0.0279
Precision	91.559%	Insight	0.9299	0.9792	0.0731
Recall	91.558%	Text2Img	0.9926	0.9988	0.0086
F1 Score	91.556%	Wiki	0.9297	0.9758	0.0783

3.4 Deepfake Defender

This model has two important components - two basemodels, RepLKNet and ConvNext. Final score is based on the scores of these two basemodels. It is obtained by merging two models, going through a softmax layer, then averaging over each batch and each frame, and finally stacking and averaging again.

3.4.1 BaseModel

RepLKNet RepLKNet comprises Stem, Stages and Transitions. Stem is the initial layer, which is used to capture low-level features. There are 4 Stages, each Stage contains multiple RepLK Blocks and ConvFFN Blocks. Transition Blocks are used to increase the number of channels in the feature map.

ConvNext Team from meta started with ResNet-50, gradually modified its architecture to incorporate modern design elements, and ended up with ConvNeXt, a fully convolutional model. This is the architecture of ConvNeXt. Compared to the ResNet-50, ResNeXt model, the team has adopted a lot of tricks and Macro and Micro design to improve the performance.

3.4.2 Rationalization of Merge

Since the model is based on the fusion of two basemodels, we believe that it is necessary to address the merge rationalization for the models, and the following are the commonalities and differences of the models:

Commonality Commonalities in models can be used as a basis for merge, improving overall robustness, and commonalities are as below:

- Both are networks obtained by improving on convolutional neural networks.
- Both models are all benchmarked against swin-transformer, which is very hot in 2021, and the overall form also references the structure of swin-transformer.
- Both use large convolutional kernels, and some similar approaches: the activation function uses gelu

Differentiation Model differentiation can further improve overall performance and generalization capabilities, and differentiations are as below:

- RepLKNet uses bn, which is more suitable for convolutional neural networks, while the ConvNeXt network takes a page from the swin-transformer idea and replaces BN with LN, and finds improved performance.
- RepLKNet uses larger convolutional kernels, while ConvNeXt has more blocks.

3.4.3 Model Evaluation

Table 4: Training results of 2 basemodels of deepfake defender model with four datasets (Four classes: SD v1.5, SD inpainting, InsightFace, Real.)

Model	Method	ACC	AUC	ERR
RepLKNet	SD v1.5	0.9974	0.9999	0.0035
	SD Inpainting	0.9950	0.9999	0.0398
	InsightFace	0.9618	0.9938	0.0398
	Real	0.9614	0.9834	0.0379
ConvNeXt	SD v1.5	0.9820	0.9991	0.0093
	SD Inpainting	0.9795	0.9994	0.0225
	InsightFace	0.9275	0.9838	0.0755
	Real	0.9170	0.9756	0.0575

4 Generalization

To evaluate the generalizability of each fine-tuned model, we tested them on the DiffusionDB-Face (DFDB) dataset [2]. Benchmarks for the DFDB dataset, as reported in the original study, are presented in Table 5:

Table 5: Performance of various models on DFF and DFDB datasets.

Model	DFF			DFDB		
	AUC	EER	ACC	AUC	EER	ACC
HiFi Net	0.57	0.45	0.45	0.52	0.66	0.51
SBI	0.51	0.61	0.50	0.25	0.89	0.30
CADD	0.51	0.49	0.50	0.48	0.70	0.47
CNNDet	0.61	0.41	0.58	0.53	0.49	0.52
DSP-FWA	0.50	0.88	0.40	0.52	0.51	0.54
Capsule	0.49	0.49	0.50	0.48	0.57	0.46
DIRE	0.38	0.76	0.55	0.62	0.45	0.71

4.1 Model Robustness: ResNet

In real-world scenarios, images are often subjected to noise from compression, filters, or other distortions. Robustness to such perturbations is crucial for reducing false positives in deepfake

detection. To evaluate the robustness of ResNet models, we introduced various perturbations to the input images, including saturation, contrast, Gaussian blur, and pixelation. Among these, contrast and pixelation had the most significant impact on classification performance, as detailed in Table 6. Saturation had a smaller effect, likely because some input images were already highly saturated.

Additionally, adversarial examples have been shown to bypass detection by denoising fake images using diffusion models [12]. To explore this vulnerability, we reconstructed all input images using a diffusion model and observed a substantial drop in accuracy, as shown in Table 7. These results suggest that ResNet models tend to learn features specific to each generator rather than generalizable patterns indicative of manipulated content.

Generalization remains a significant challenge for CNN-based models, as also noted by Luo et al. [11]. To further assess this, we evaluated ResNet models on the DFDB dataset. As shown in Table 8, the accuracy dropped to approximately 60%, with ResNet50 achieving the highest performance at 67.01%.

Table 6: ResNet on perturbed DFF data. Saturation factor: 0.5; Contrast factor 0.5; Gaussian blur radius: 1; Pixelation downscale: 0.1. (Four classes: SD v1.5, SD inpainting, InsightFace, Real.)

Model	Method	ACC	AUC	EER
ResNet50	Saturation	0.9442	0.9628	0.0186
	Contrast	0.5825	0.7216	0.1392
	Gaussian Blur	0.7406	0.8271	0.0864
	Pixelation	0.3904	0.5936	0.2031
ResNet101	Saturation	0.9718	0.9812	0.0094
	Contrast	0.8505	0.9003	0.0499
	Gaussian Blur	0.8496	0.8997	0.0501
	Pixelation	0.3396	0.5597	0.2201
ResNet152	Saturation	0.9453	0.9635	0.0182
	Contrast	0.6636	0.7757	0.1121
	Gaussian Blur	0.8006	0.8671	0.0664
	Pixelation	0.5238	0.6826	0.1587

Table 7: ResNet on denoised DFF data. Diffusion timesteps: 50. (Four classes: SD v1.5, SD inpainting, InsightFace, Real.)

Model	ACC	AUC	EER
ResNet50	0.2500	0.5000	0.0000
ResNet101	0.2500	0.5000	0.0000
ResNet152	0.2298	0.4865	0.5673

Table 8: ResNet on DFDB. (Two classes: Real, Fake.)

Model	ACC	AUC	EER
ResNet50	0.6701	0.7293	0.3179
ResNet101	0.6147	0.6317	0.3976
ResNet152	0.6056	0.6340	0.4014

4.2 Model Robustness: EfficientNet

EfficientNet models were also tested on the **DFDB** dataset for binary classification (Real vs. Fake). As shown in Table 9, EfficientNet-B2 achieved high accuracy on the training dataset but performed poorly on unseen data, indicating limited generalization capabilities. Metrics such as precision and AUC also reveal a high false positive rate and near-random classification for certain classes.

4.3 Model Robustness: Vision Transformer (ViT)

The Vision Transformer (ViT) was evaluated for generalization using the **DFDB** and **FFHQ** (Flickr-Faces-HQ) dataset. Table 10 shows that while the ViT model maintained reasonable accuracy on the original test set, its performance degraded on unseen datasets, highlighting challenges in adapting to new generative techniques.

Table 9: EfficientNet-B2 on DiffusionDB-Face

Metric	Value
Accuracy	0.4161
Precision	0.4167
Recall	0.9957
ROC AUC	0.4980
Equal Error Rate (EER)	0.9997

Table 10: ViT on DFDB and FFHQ

Metric	Value
Accuracy	0.8878
Precision	0.4277
Recall	0.5969
F1 Score	0.4983
AUC	0.8579
EER	0.2214

4.4 Model Robustness: ResNet50-ViT Hybrid

The ResNet50-ViT hybrid model integrates the ResNet50 backbone for feature extraction with a Vision Transformer (ViT) for classification, aiming to combine the hierarchical feature extraction capabilities of ResNet50 with the long-range dependencies learned by ViT. During training, ResNet50 processes the input images to produce feature maps, which are then passed to the ViT model for further processing and final classification. While the model demonstrated strong performance on the DFDB dataset (Table 11), achieving high accuracy and AUC, its performance significantly degraded on the Diffusion DB+thumbnails dataset (Table 12), revealing challenges in generalizing to unseen data with different generative techniques.

Table 11: ResNet50-ViT on DFDB (Training Results)

Metric	Value
Accuracy	0.9841
AUC	0.9895
EER	0.0053
Class 0 Accuracy	0.9997
Class 1 Accuracy	0.9993
Class 2 Accuracy	0.9844
Class 3 Accuracy	0.9847

Table 12: ResNet50-ViT on Diffusion DB+Thumbnails (Test Results)

Metric	Value
Accuracy	0.6903
AUC	0.6292
EER	0.0000

4.5 Adversarial Training for Improved Generalization

The generalization challenges observed in CNN-based models, such as ResNet and EfficientNet, align with findings in Section 2, highlighting their susceptibility to overfitting and limited ability to integrate global features for robust detection. To mitigate these issues, we conducted adversarial training on ResNet50, which demonstrated the highest accuracy on the DiffusionDB-Face (DFDB) dataset among the tested ResNet models.

To improve the model’s robustness, we introduced perturbations, including contrast adjustments, pixelation, and Gaussian noise. Gaussian noise was selected due to its relevance to diffusion models. As shown in Table 13, adversarial training with pixelation significantly enhanced the model’s performance on unseen data. Although pixelation reduced the model’s accuracy on the DeepFakeFace (DFF) dataset from 90.22% to 83.98%, it increased accuracy on the DFDB dataset from 67.01% to 78.05%, demonstrating improved generalization.

However, combining both contrast adjustments and pixelation in a single training pipeline did not yield further improvements. The accuracy on the DFF dataset decreased slightly to 83.90%, and the accuracy on the DFDB dataset dropped to 73.79%, falling short of the gains achieved with pixelation alone. This suggests that while adversarial training with individual perturbations like pixelation can enhance robustness, combining multiple perturbations may introduce competing effects that hinder overall performance.

Table 13: Adversarial training (Fine-tuned ResNet50)

Test Dataset	Method	ACC	AUC	EER
Deepfake Face	Contrast	0.9183	0.8601	0.2567
	Pixelation	0.8392	0.7241	0.5065
	Gaussian noise	0.7661	0.5655	0.8372
DiffusionDB-Face	Contrast	0.6703	0.7736	0.2851
	Pixelation	0.7805	0.8732	0.2059
	Gaussian noise	0.4568	0.4623	0.5283

5 Conclusion and Future Work

This study evaluated multiple models for deepfake detection, including ResNet, EfficientNet, Vision Transformers, and Deepfake Defender, to identify manipulated content and analyze the vulnerabilities of AI systems to various deepfake generation techniques. The models demonstrated strong performance on curated datasets, with fine-tuning and performance enhancement techniques yielding incremental improvements over their robust baseline capabilities. These findings underline the effectiveness of state-of-the-art architectures in tackling deepfake detection challenges.

To further assess their real-world applicability, we tested the generalization of these models on different datasets. This analysis revealed critical limitations, particularly when models encountered unseen generative techniques or adversarial conditions. These results underscore the importance of addressing overfitting and enhancing model adaptability to improve robustness in diverse scenarios. The insights gained from this study provide valuable guidance for researchers and practitioners working to advance deepfake detection technologies.

Future Work:

- **Enhanced Generalization Techniques:** Explore advanced training strategies, such as self-supervised learning or domain adaptation, to improve model robustness on unseen data.
- **Dataset Diversity:** Expand the scope of datasets to include a broader range of generative techniques and real-world adversarial scenarios for more comprehensive evaluations.
- **Adversarial Defense Mechanisms:** Investigate hybrid approaches that combine adversarial training techniques with CNNs and Transformer models to enhance resilience against evolving generative methods.
- **Explainability and Trust:** Incorporate explainable AI techniques to improve transparency and trust by elucidating model decisions, especially in critical use cases.

By addressing these areas, future research can build on the foundations laid by this study to create more adaptable, reliable, and explainable deepfake detection systems capable of meeting the challenges posed by rapidly evolving generative AI technologies.

References

- [1] Song, H., Huang, S., Dong, Y., & Tu, W. W. (2023). Robustness and generalizability of deepfake detection: A study with diffusion models. arXiv preprint arXiv:2309.02218.
- [2] Bhattacharyya, C., Wang, H., Zhang, F., Kim, S., & Zhu, X. (2024). Diffusion deepfake. arXiv preprint arXiv:2404.01579.
- [3] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (pp. 770–778).
- [4] Wang, S. Y., Wang, O., Zhang, R., Owens, A., & Efros, A. A. (2020). CNN-generated images are surprisingly easy to spot... for now. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 8695–8704).

- [5] Tan, M., & Le, Q. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. In International Conference on Machine Learning (pp. 6105–6114). PMLR.
- [6] Agarwal, S., Farid, H., Fried, O., & Milborrow, J. (2020). Detecting deep-fake videos from phoneme-viseme mismatches. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (pp. 660–661).
- [7] Coccomini, D. A., Messina, N., Gennaro, C., & Falchi, F. (2022). Combining EfficientNet and Vision Transformers for video deepfake detection. In International Conference on Image Analysis and Processing (pp. 219–229). Springer International Publishing.
- [8] Heo, Y. J., Yeo, W. H., & Kim, B. G. (2023). Deepfake detection algorithm based on improved vision transformer. *Applied Intelligence*, 53(7), 7512–7527.
- [9] Wang, Z., Bao, J., Zhou, W., Wang, W., Hu, H., Chen, H., & Li, H. (2023). DIRE for diffusion-generated image detection. In Proceedings of the IEEE/CVF International Conference on Computer Vision (pp. 22445–22455).
- [10] Alexey, D. (2020). An image is worth 16x16 words: Transformers for image recognition at scale. arXiv preprint arXiv:2010.11929.
- [11] Luo, Y., Zhang, Y., Yan, J., Liu, W. (2021). Generalizing face forgery detection with high-frequency features. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 16317–16326).
- [12] Ivanovska, M., & Struc, V. (2024). On the vulnerability of deepfake detectors to attacks generated by denoising diffusion models. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (pp. 1051–1060).